



اَوْنُوْ رَسِيْئِيْ رَاتِيْكَوْ لَوِيْ مَارَا
UNIVERSITI
TEKNOLOGI
MARA

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES

Universiti Teknologi MARA (UiTM) Shah Alam

BACHELOR OF INFORMATION SYSTEM (HONS.)

INFORMATION SYSTEM ENGINEERING

CSC584 – ENTERPRISE PROGRAMMING

TITLE:

E-SUKARELAWAN: VOLUNTEER HOURS & SOCIAL CREDIT

PREPARED BY:

NAME	STUDENT ID
GLORIA TRIZA MOJIUN	2024803882
NURUL ATIRAH BINTI MOHD YUZI	2024634238
AYUNI FARHANA BINTI ROSLI	2024234738
NURUL IMAN IZZATI BINTI MOHD AMHARI	2024274376

GROUP: CDCS2664A

PREPARED FOR:

SIR MUHAMMAD RIDHWAN MOHAMAD RAZALI

DATE OF SUBMISSION: 6/7/2026

Table of Contents

1.	INTRODUCTION	3
2.	PROBLEM STATEMENT	4
3.	OBJECTIVE	5
4.	ERD	6
5.	USER MANUAL	7
6.	CONCLUSION.....	14

1. INTRODUCTION

The E-Sukarelawan: Volunteer Hours and Social Credit System is a web-based platform designed to manage volunteer activities and community service hours at the university. This system uses Java Web Technology, the Model-View-Controller (MVC) architecture and the Apache Derby database, allowing for an efficient and centralized solution for volunteer management. Organizers can create and manage volunteer calls, record attendance and monitor participants, while volunteers can browse volunteering opportunities, join activities and track their total service hours. By digitalizing the volunteer management process, E-Sukarelawan improves organization, transparency, and encourages greater student participation in community service activities.

2. PROBLEM STATEMENT

Many universities encourage students to participate in volunteer and Corporate Social Responsibility (CSR) activities to promote community engagement and personal development. However, the management of volunteer activities is often handled manually through social media, messaging applications, or paper-based records. This approach creates several challenges for students, club leaders, and university administrators.

Firstly, students often face difficulties in finding reliable and updated information about available volunteering opportunities. Announcements are scattered across different platforms, causing students to miss important activities and registration deadlines.

Secondly, the recording of volunteer participation and service hours is usually done manually. This process is time-consuming and prone to errors, making it difficult for students to monitor their accumulated volunteer hours and achievements accurately.

In addition, club leaders and NGOs experience challenges in managing volunteer calls, tracking participant records, and updating activity information efficiently. Manual record management increases administrative workload and reduces data accuracy.

Furthermore, universities lack a centralized platform to monitor the overall community service contributions made by students and student organizations. Without a proper system for recording and displaying volunteer participation and service hours, it is difficult to monitor students' contributions, manage volunteer records efficiently, and obtain an overview of the university's community service activities.

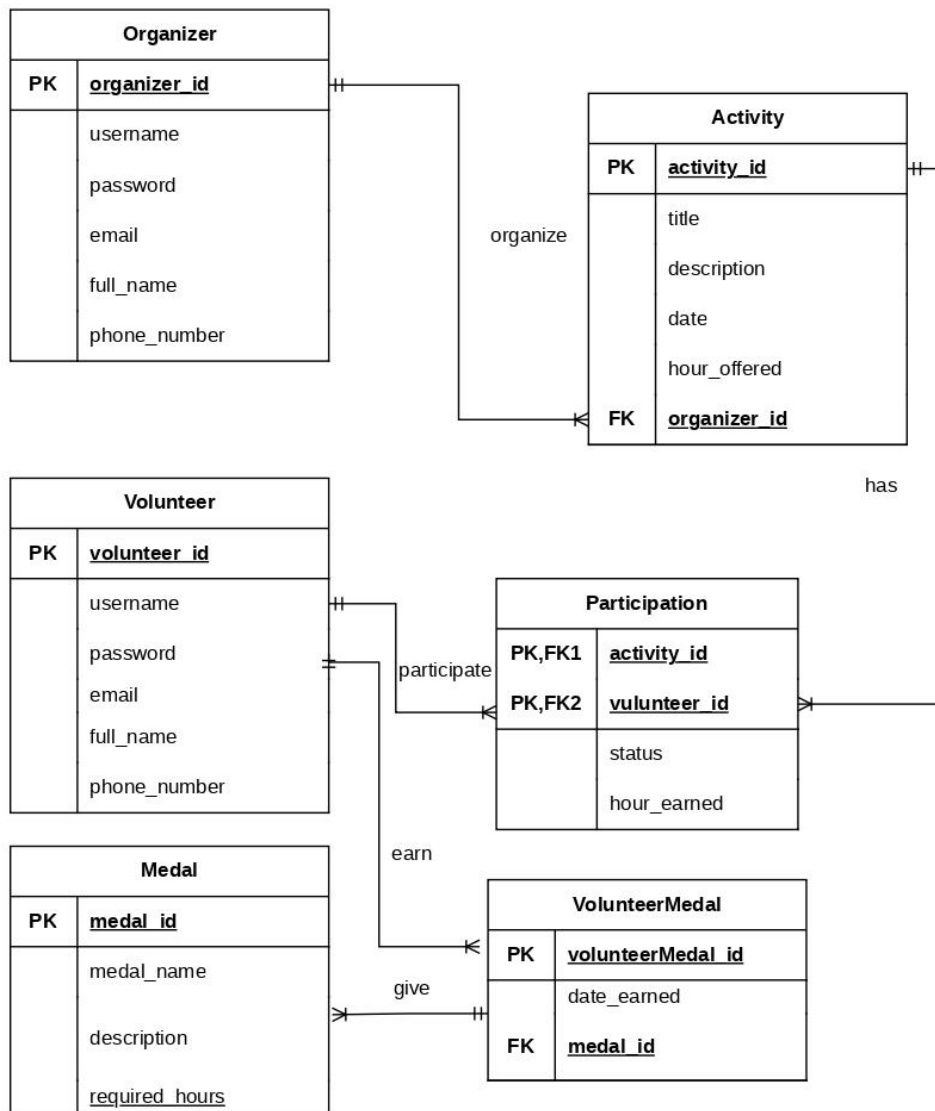
Therefore, the E-Sukarelawan: Volunteer Hours & Social Credit System is proposed to provide a centralized web-based platform that enables efficient volunteer activity management, attendance recording, and participation tracking for students, club leaders, and university administrators.

3. OBJECTIVE

The objectives of the E-Sukarelawan: Volunteer Hours & Social Credit System are:

1. To develop a web-based platform that allows students to easily discover and participate in volunteer and CSR activities organized by NGOs and student clubs.
2. To provide a centralized system for recording, tracking, and managing students' volunteer hours and participation history accurately.
3. To enable NGOs and club leaders to efficiently create, update, view, and delete volunteer calls through an online management system.
4. To provide an interactive dashboard that displays the total community service hours contributed by students, volunteer participation records, and activity statistics to help organizers monitor volunteer activities effectively.

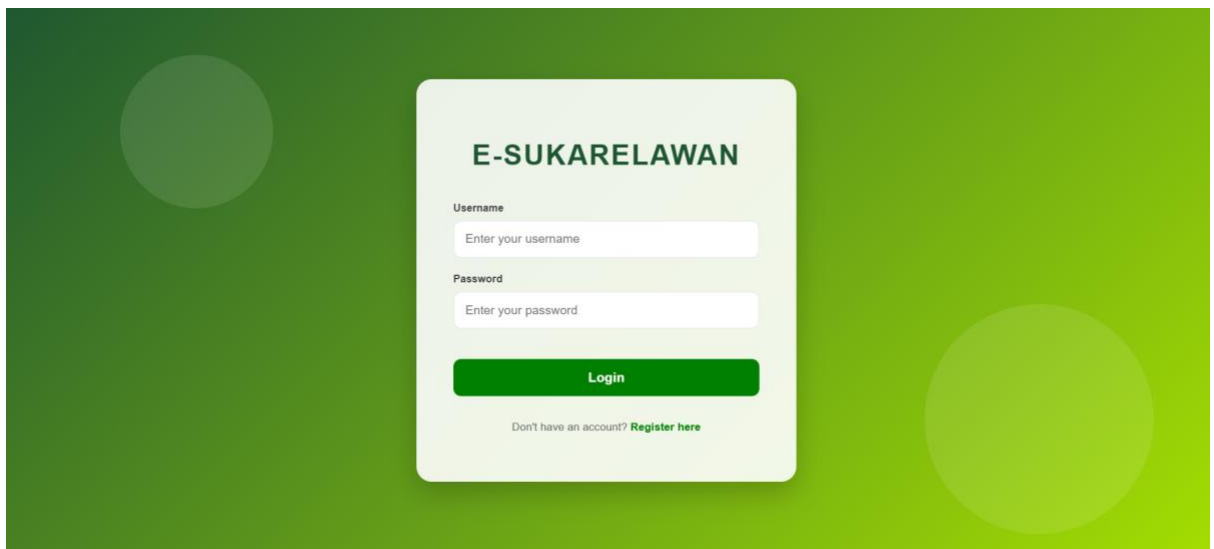
4. ERD



5. USER MANUAL

i. Login Page

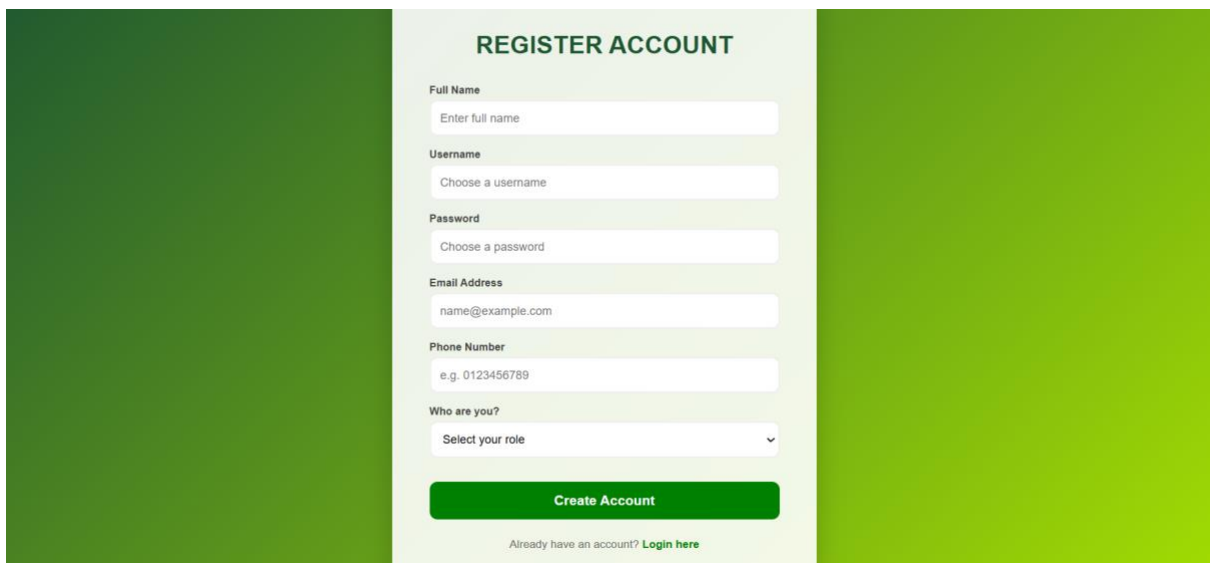
- Enter your username and password.
- Click Login to access the system.
- Click Register Here if you do not have an account.



The login page features a green gradient background with two large, faint circular shapes. A central white card contains the title "E-SUKARELAWAN" in bold green text. Below the title are two input fields: "Username" with the placeholder "Enter your username" and "Password" with the placeholder "Enter your password". A green "Login" button is positioned below the password field. At the bottom of the card, a link reads "Don't have an account? [Register here](#)".

ii. Register Page

- Fill in all your personal information.
- Choose your role, either Student (Volunteer) or NGO/Club Leader.
- Click Create Account to register a new account.
- After successful registration, you will be redirected to the Login Page.



The registration page has a green gradient background. A central white card is titled "REGISTER ACCOUNT" in bold green text. It contains several input fields: "Full Name" (placeholder: "Enter full name"), "Username" (placeholder: "Choose a username"), "Password" (placeholder: "Choose a password"), "Email Address" (placeholder: "name@example.com"), and "Phone Number" (placeholder: "e.g. 0123456789"). Below these is a dropdown menu labeled "Who are you?" with the option "Select your role" and a downward arrow. A green "Create Account" button is at the bottom of the card. A link at the very bottom reads "Already have an account? [Login here](#)".

iii. Dashboard (Organizer)

- View Total Activities, Community Hours and Total Volunteers.
- Search volunteer activities using the search box.
- Click the notification icon to view students who have recently joined volunteer activities. These notifications remain until the organizer records their attendance.
- Manage volunteer calls using the Attendance, Edit and Delete buttons.
- View the Student Participation section to monitor students who have completed volunteer activities, including their activity name, participation status, and hours earned.

The screenshot shows the E-SUKARELAWAN Dashboard for an Organizer named Atirah. The dashboard includes a sidebar with navigation links for Dashboard, Volunteer Calls, and Participation. The main content area features a welcome message, summary cards for Total Activities (1), Community Hours (2), and Total Volunteers (1), a Volunteer Calls Management table with a 'Tree Planting Campaign' entry, and a Student Participation table showing the same campaign as completed.

E-SUKARELAWAN Dashboard [Logout](#)

Welcome, Atirah!
Here's an overview of your volunteer activities and participation.

1
Total Activities

2
Community Hours

1
Total Volunteers

Volunteer Calls Management [+ Create Volunteer Call](#)

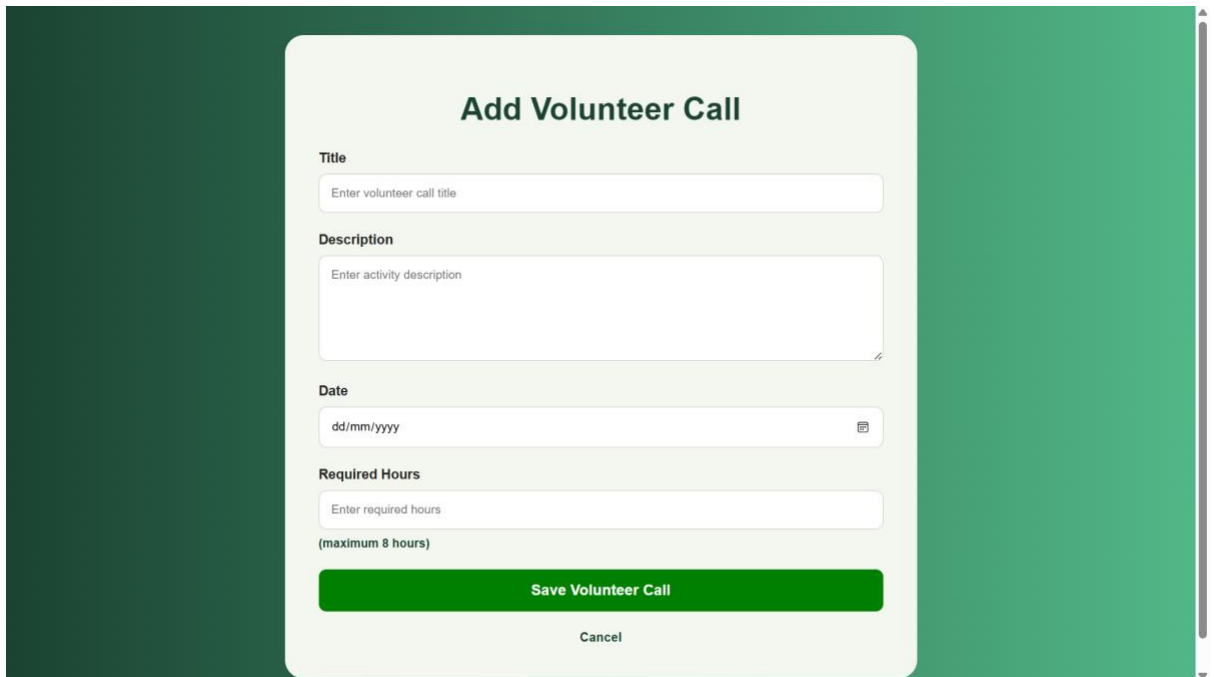
Title	Description	Date	Hours	Organizer	Actions
Tree Planting Campaign	Participate in a tree planting programme to promote environmental sustainability. Volunteers will help plant and maintain trees around the community park.	2026-07-23	2	Atirah	Attendance Edit Delete

Student Participation

Activity	Total Participants	Status	Hours Earned	Actions
Tree Planting Campaign	1	Completed	2	View Details

iv. Create Volunteer Call (Organizer)

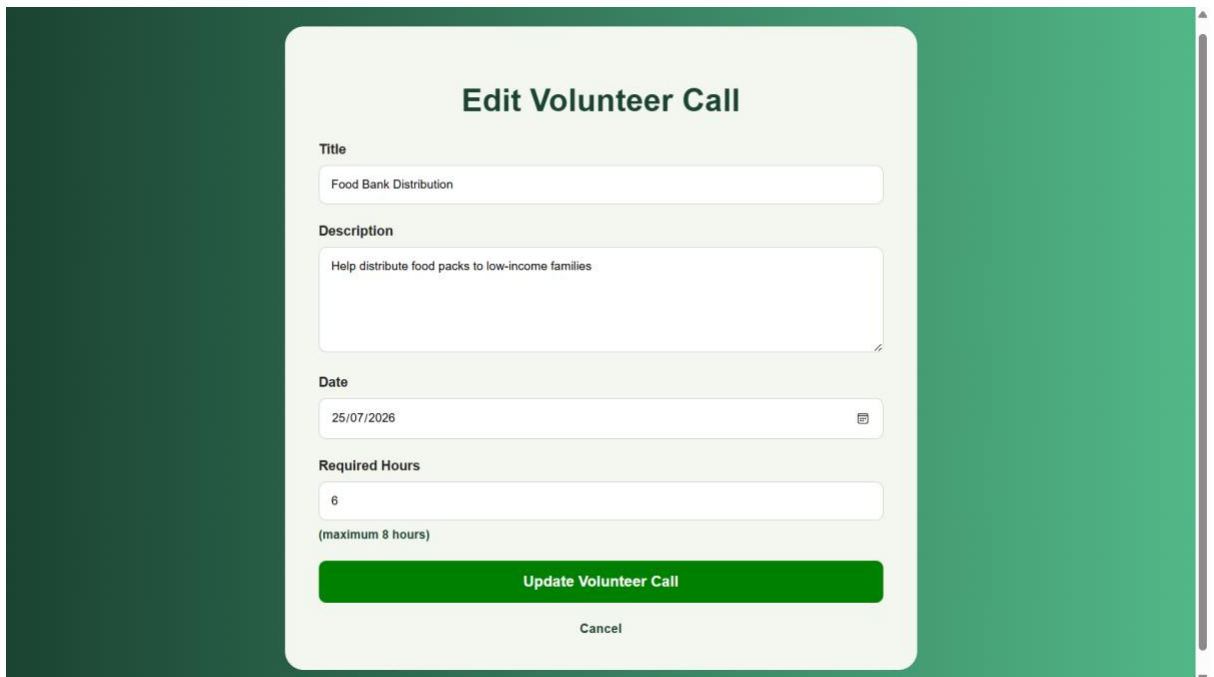
- Fill in the title, description, date and required hours.
- Click Save Volunteer Call.



The screenshot shows a web form titled "Add Volunteer Call" on a green background. The form is a light green rounded rectangle with the following fields: "Title" with a placeholder "Enter volunteer call title", "Description" with a placeholder "Enter activity description" and a text area, "Date" with a placeholder "dd/mm/yyyy" and a calendar icon, and "Required Hours" with a placeholder "Enter required hours" and a note "(maximum 8 hours)". At the bottom are two buttons: a green "Save Volunteer Call" button and a grey "Cancel" button.

v. Edit Volunteer Call (Organizer)

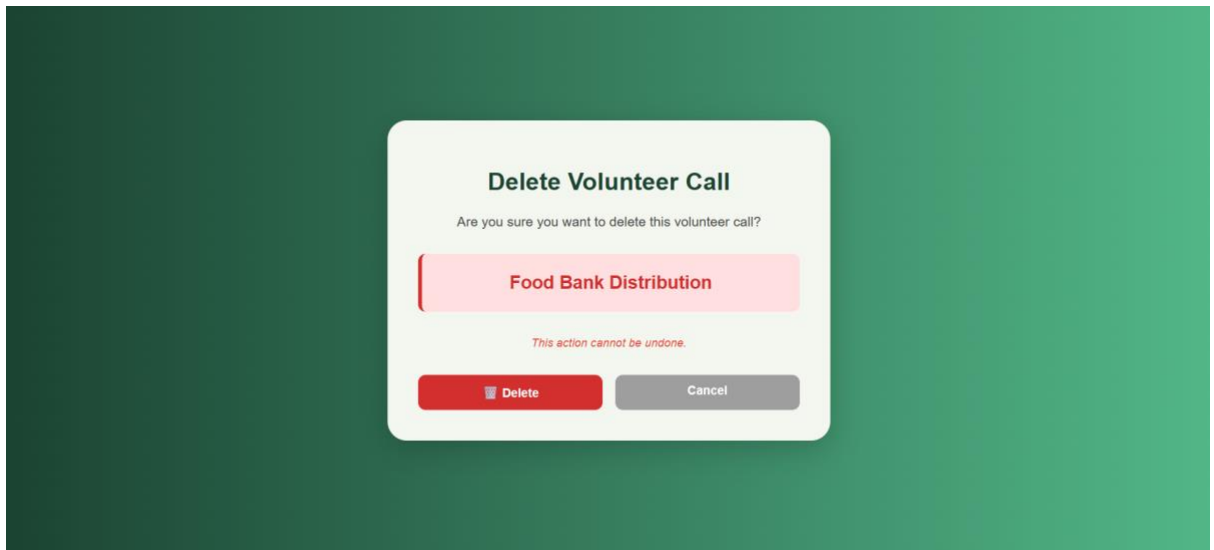
- Modify the volunteer call information.
- Click Update Volunteer Call to save the changes.



The screenshot shows a web form titled "Edit Volunteer Call" on a green background. The form is a light green rounded rectangle with the following fields: "Title" with the value "Food Bank Distribution", "Description" with the value "Help distribute food packs to low-income families" and a text area, "Date" with the value "25/07/2026" and a calendar icon, and "Required Hours" with the value "6" and a note "(maximum 8 hours)". At the bottom are two buttons: a green "Update Volunteer Call" button and a grey "Cancel" button.

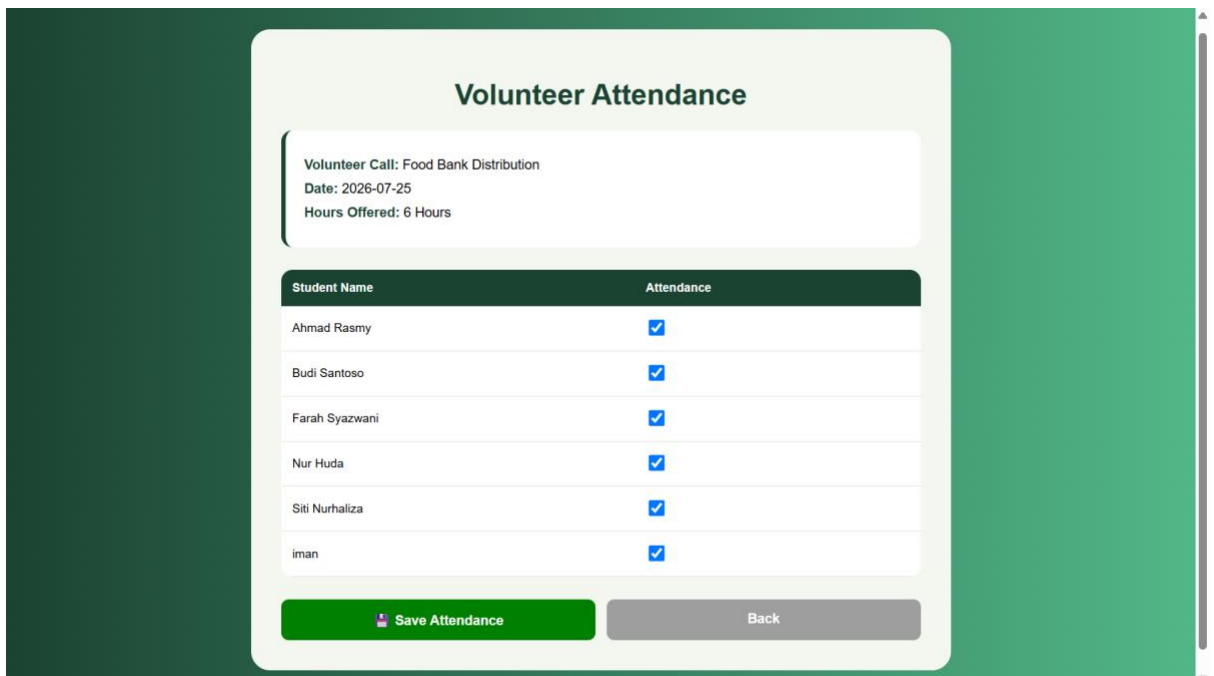
vi. Delete Volunteer Call (Organizer)

- Click the Delete button from Volunteer Calls Management table.
- Confirm the deletion.



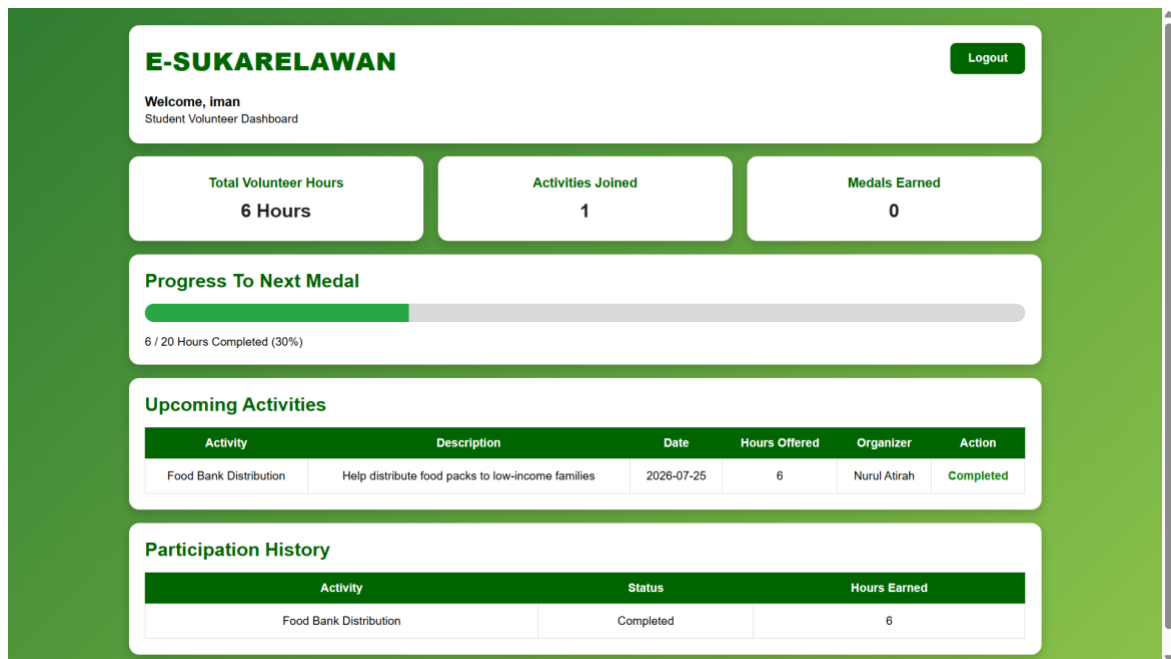
vii. Attendance Management (Organizer)

- View students who joined the activity.
- Tick the attendance checkbox.
- Click Save Attendance.



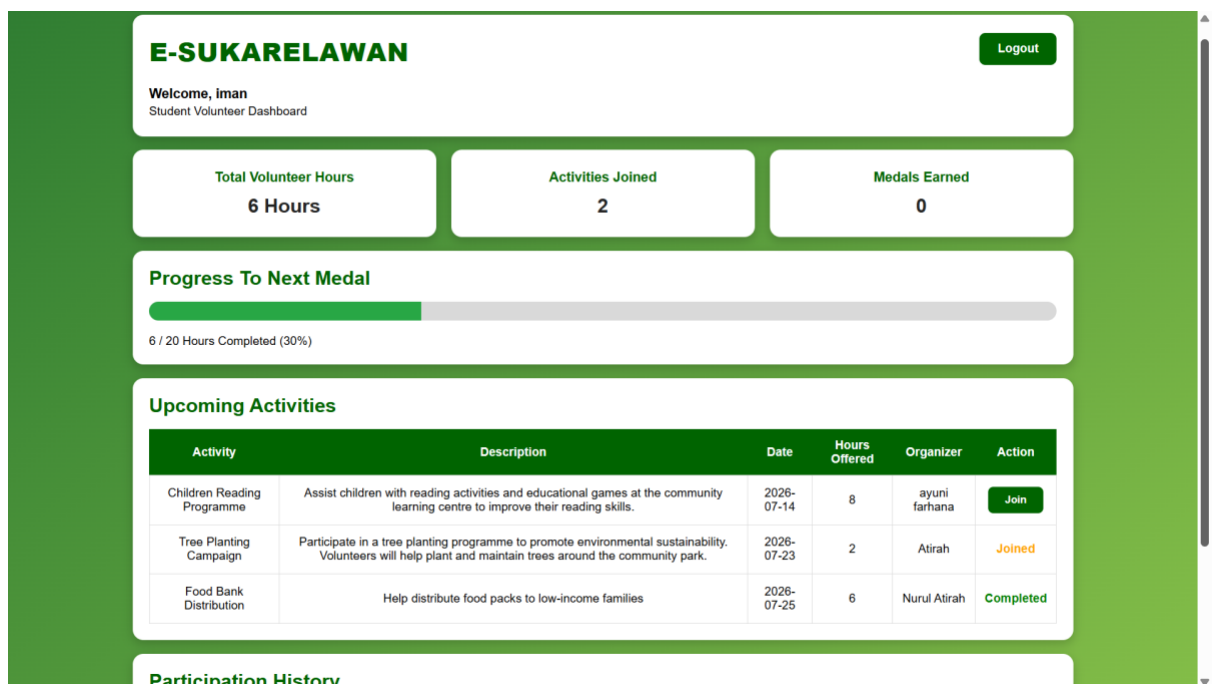
viii. Dashboard (Student)

- View Total Volunteer Hours, Activities Joined and Medals Earned.
- View Progress To Next Medal, Upcoming Activities and Participation History.



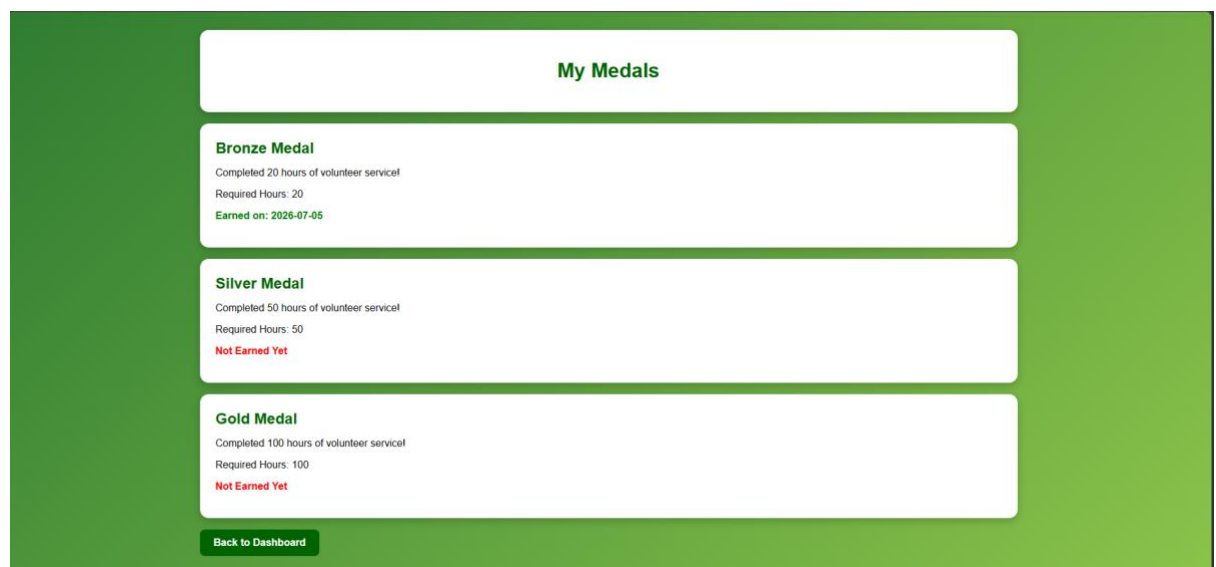
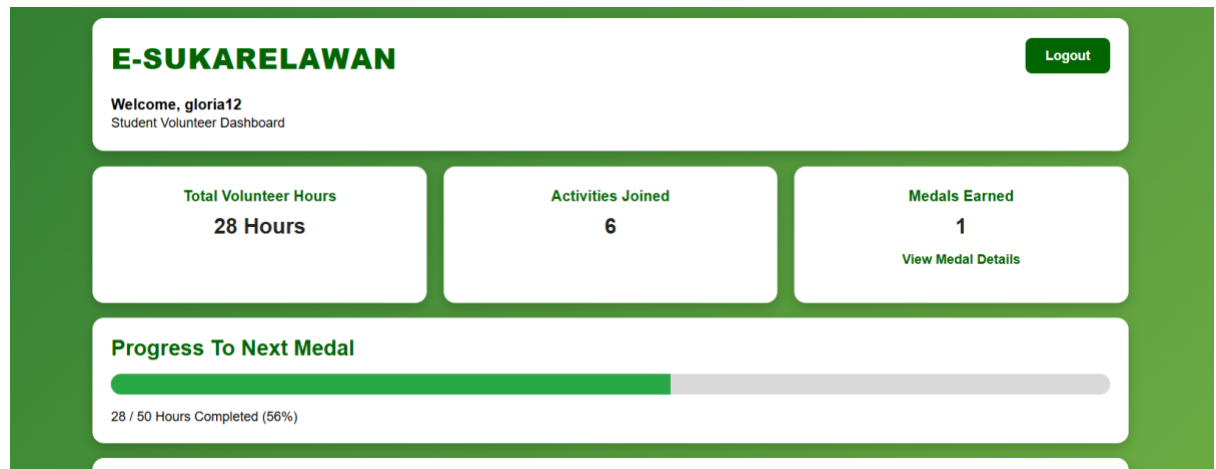
ix. Join Volunteer Activity (Student)

- Click Join beside an available volunteer activity.
- The activity status will change to Joined after you click Join.
- Once the organizer records your attendance, the status will automatically change to Completed, and your volunteer hours will be updated.



x. Medal Earned

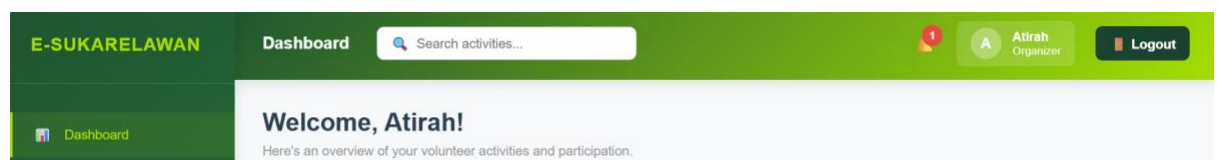
- Click Medal Earned to see medal achievement



xi. Logout

- Click the Logout button at the top right corner of the dashboard.
- You will be logged out of the system and redirected to the Login Page.

(Organizer Dashboard)



(Student Volunteer dashboard)

E-SUKARELAWAN

Logout

Welcome, iman
Student Volunteer Dashboard

6. CONCLUSION

In conclusion, the E-Sukarelawan: Volunteer Hours & Social Credit System provides a more systematic and efficient way to manage volunteer activities and community service hours. By using Java Web Technology, the Model-View-Controller (MVC) architecture, and the Apache Derby database, the system provides a reliable and user-friendly platform for both organizers and students. This platform makes it easier for organizers to manage volunteer calls and attendance, and allows students to easily browse available activities and track their participation records. The system successfully manages volunteer activities, encourages more student participation in community service, and provides a centralized platform for volunteer management in the university.